

GeoLokal Web GIS System Design Specification

I. Executive Summary

This report presents the planning, architecture, and technical design for the proposed Web Geographic Information System (Web GIS). The system is intended to serve as a centralized, web-based platform for managing, visualizing, analyzing, and sharing geospatial information to support planning, operations, and decision-making.

The Web GIS will enable authorized users to access up-to-date spatial data through interactive maps, dashboards, and reports. It is designed to improve data accessibility, enhance inter-departmental coordination, and support evidence-based decision-making. This report provides both an executive-level overview for decision-makers and a technical design specification for system implementation.

II. Objective and Scope

The primary objectives of the GeoLokal Web GIS are to:

- Centralize spatial and non-spatial data in a secure, web-accessible platform for all LGUs
- Enable interactive visualization of geospatial data
- Support planning, monitoring, and reporting functions
- Improve efficiency and transparency in spatial decision-making
- Build institutional capacity through training and knowledge transfer

The scope of the GeoLokal Web GIS includes:

- Web-based map visualization and navigation, providing interactive access to spatial data through standard browser-based tools, including map navigation, layer control, and data exploration features.
- Role-based access control, ensuring that only authorized users can view, edit, or administer system components and datasets based on assigned roles and LGU-level permissions.
- Configurable dashboards and reporting pages, tailored to the specific needs of each city or municipality, enabling localized performance monitoring, analytics, and evidence-based decision-making.
- System administration and data management tools, supporting user management, dataset configuration, access control, auditing, and routine system maintenance.

III. Stakeholders & Users

The successful implementation of the GeoLokal Web GIS requires coordination among multiple stakeholders, including:

- System owner management
- System administrators and IT personnel
- End users (LGUs)

A governance structure will be established to define decision-making authority, data ownership, system approvals, and change management processes. Clear roles and responsibilities are essential to ensure sustainability and accountability.

IV. Functional Requirements

The system shall provide the following functional capabilities:

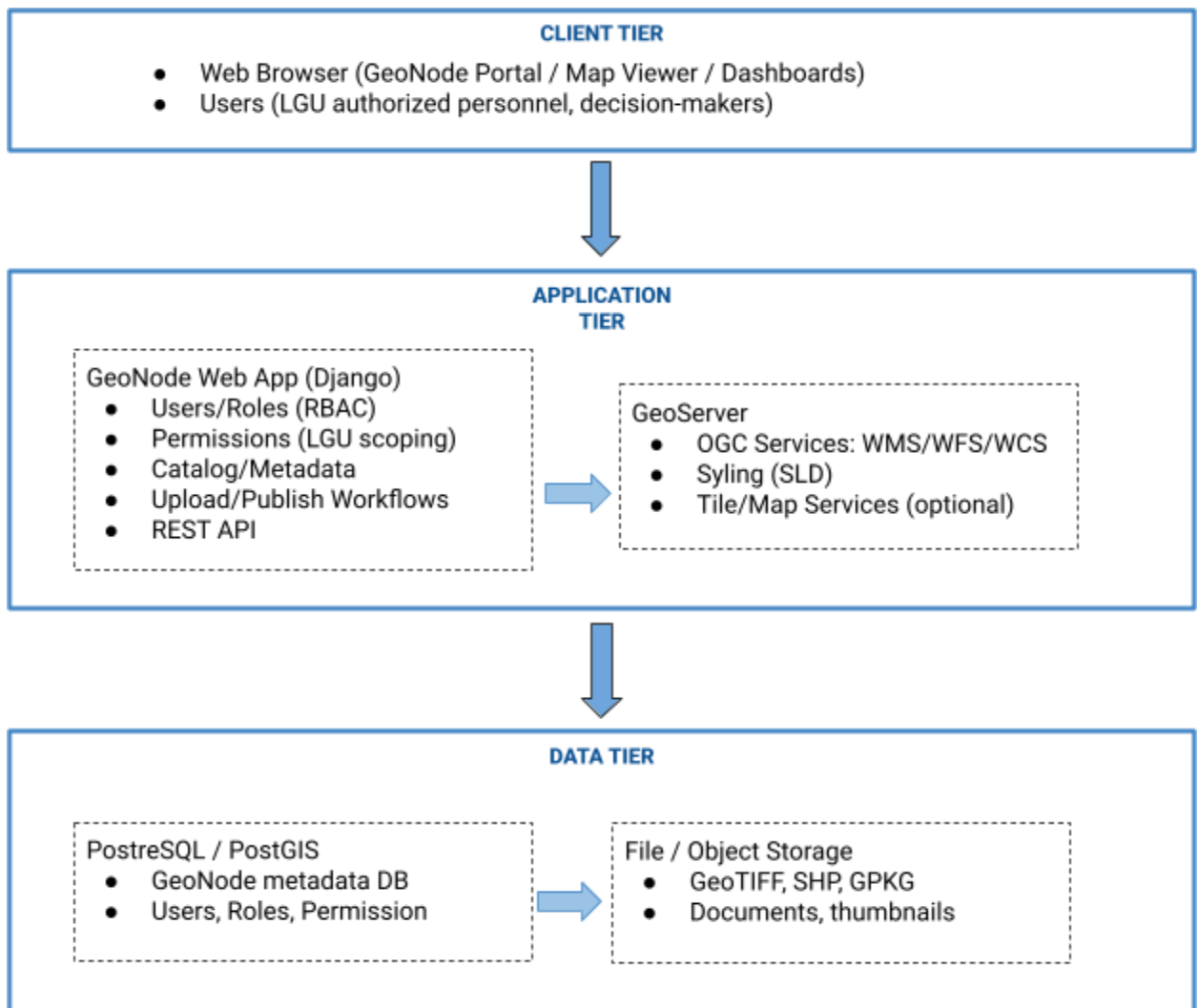
- Web-based map visualization and navigation, providing interactive access to spatial data through standard browser-based tools, including map navigation, layer control, and data exploration features.
- Role-based access control, ensuring that only authorized users can view, edit, or administer system components and datasets based on assigned roles and LGU-level permissions.
- Configurable dashboards and reporting pages, tailored to the specific needs of each city or municipality, enabling localized performance monitoring, analytics, and evidence-based decision-making.
- System administration and data management tools, supporting user management, dataset configuration, access control, auditing, and routine system maintenance.

V. Non-Functional Requirements

The system shall meet the following non-functional requirements:

- Adequate performance for concurrent users
- High availability and reliability
- Scalability to accommodate future data and users
- Secure access and data protection
- Usable interface accessible via modern web browsers

VI. System Architecture



The Web GIS follows a three-tier architecture implemented using GeoNode as the core platform:

Client Tier

End-user interfaces used to access maps, layers, dashboards, and admin tools

- Web browsers (desktop/mobile)
- User experience: GeoNode UI (catalog, maps, documents), map viewer/editor
- (Optional) internal users over VPN / external users over HTTPS

Application Tier

GeoNode services that handle authentication, permissions, publishing, and map services

- Nginx (Reverse Proxy / HTTPS termination)
 - Routes traffic, enforces HTTPS, and optional rate-limiting

- GeoNode Web App (Django + GeoNode APIs)
 - Handles portal pages, users/roles, permissions, metadata/catalog, uploads, workflows
- GeoServer (OGC Services)
 - Publishes spatial layers through WMS/WFS/WCS, styling via SLD, tile services if enabled

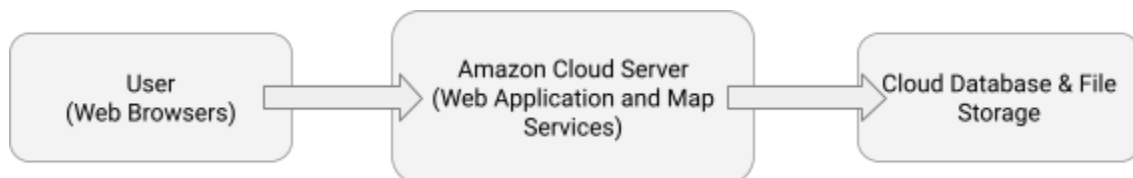
Data Tier

Where GeoNode stores authoritative data

- PostgreSQL/PostGIS
 - GeoNode metadata DB (users, permissions, layer metadata, documents)
 - Spatial feature storage for vector layers (if configured)
- File/Object Storage (local disk or S3-compatible)
 - Stores uploaded files: GeoTIFFs, Shapefiles, GeoPackages, documents, thumbnails
- Backups / Digital Twin

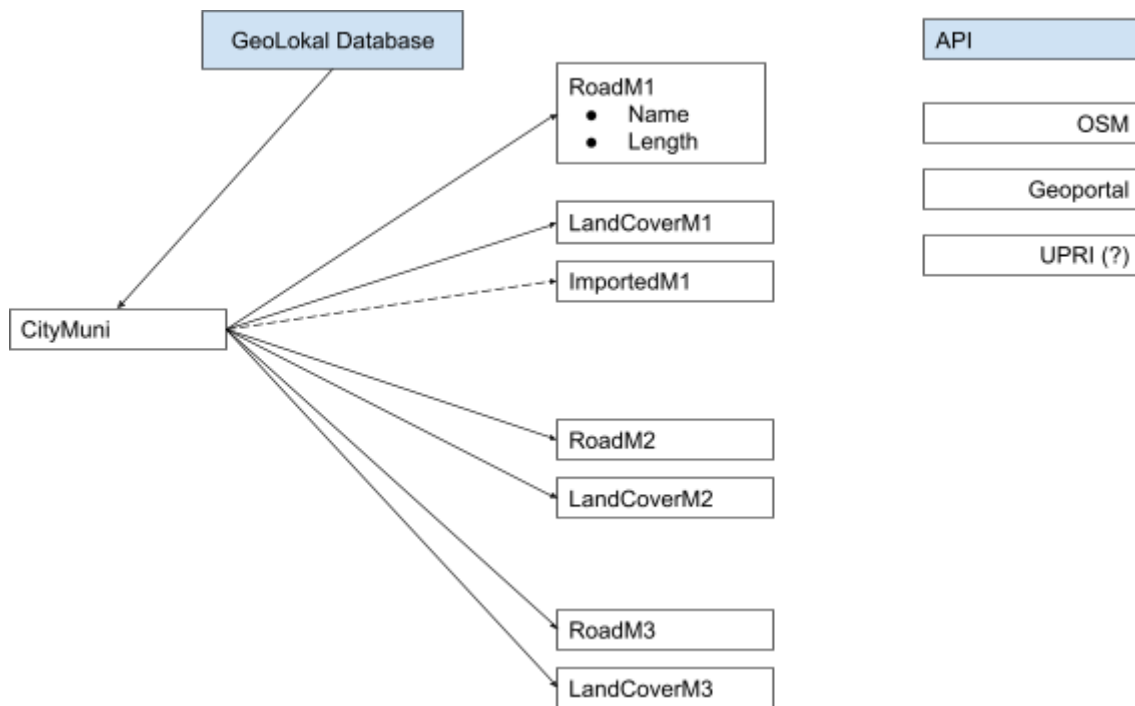
This architecture separates user access, application services, and data storage—improving security boundaries, scaling options, and maintainability.

Cloud Server Deployment Diagram



This diagram illustrates the Web GIS deployed on Amazon Cloud, where users access the system via the internet, application services run on cloud servers, and data is stored securely in cloud-based databases and storage.

VII. Database Management Design



The database is designed to support multiple LGUs in the future using a centralized architecture. There will be a single database with one Municipality/City master table that stores the city or municipality name and its core attributes. This table will serve as the primary reference for all related spatial and non-spatial layers within each LGU, such as roads, building footprints, land use, and any other datasets provided by the LGU.

Each LGU will also have the capability to upload and manage its own layers, which will be linked to their corresponding municipality or city record through defined relationships.

These external systems are not directly connected to the core database, the platform can integrate free government or official APIs to pull in additional information (e.g., socio-economic data, base maps, or national datasets) to enrich the LGU data and provide broader context for analysis and decision-making.

VIII. Technology Stack

The Web GIS will be implemented using a modern, open-source and cloud-based technology stack deployed on Amazon Cloud, designed for scalability, security, and maintainability.

1. Cloud Infrastructure
 - Amazon Cloud (AWS) – hosting environment for application servers, databases, and storage services
 - Cloud-based virtual servers for running web and map services
 - Cloud storage services for spatial data files and documents
2. Application Layer
 - Web GIS platform: GeoNode
 - Provides the web portal, user management, role-based access control, data catalog, and publishing workflows
 - Map services engine: GeoServer
 - Serves spatial data through standard web map services
 - Web server / reverse proxy: Nginx
 - Handles web traffic and secure connections
3. Frontend (User Interface)
 - Web-based user interface accessible via modern web browsers
 - Interactive map viewer for navigation, layer control, and data exploration
 - Dashboard and reporting interface for KPIs and summaries
4. Data Layer
 - Spatial database: PostgreSQL with PostGIS
 - Stores spatial and non-spatial datasets, metadata, and system configurations
 - Cloud file storage: Object storage for raster files, uploaded datasets, documents, and backups
5. DevOps, Source Control, and Project Monitoring
 - Source Code Management: GitHub
 - Used for version control, collaboration, issue tracking, and technical documentation
 - Agile Project Management: Trello
 - The project will be implemented using an agile approach, allowing incremental delivery, continuous feedback, and regular validation with stakeholders.
 - Development work will be organized into **time-bound sprints**, with backlog refinement and sprint reviews conducted regularly.

- Used to manage the project using an agile approach, where: Project tasks are organized into product backlogs, Work is planned and executed in sprints, Sprint progress and deliverables are tracked transparently using Trello boards
 - Trello boards will provide visibility on task status, priorities, and completion throughout the project lifecycle
6. Security and Access Control
- Role-Based Access Control (RBAC) is implemented within the Web GIS platform
 - Secure communication using HTTPS
 - Audit logging for system and user activities
7. Integration and Standards
- Open Geospatial Consortium (OGC) standards (WMS, WFS, WCS)
 - REST APIs for system integration and data access
8. Operations and Monitoring
- Cloud-based monitoring and logging for system health and performance
Backup and recovery mechanisms for databases and file storage

IX.

X. User Journey Stories

This section describes how different types of users interact with the Web GIS system from login to task completion. The user journeys illustrate typical workflows based on assigned roles and access levels, highlighting how the system supports secure, role-based access to maps, dashboards, and administrative functions. These journeys ensure clarity on system usage, access boundaries, and expected user experience.

A. User Journey Story 1: LGU End User (Viewer Role)

Step	User Action	System Response
1	Accesses the Web GIS via GeoLokal Gateway using a web browser	Displays the login page
2	Logs in using assigned credentials	Authenticates user and applies LGU-based permissions
3	Lands on the main map interface	Loads interactive map with datasets limited to the user's assigned city/municipality
4	Navigates the map (pan, zoom, toggle layers)	Updates map view dynamically
5	Applies barangay filter (if needed)	Filters visible layers and dashboard summaries by selected barangay
6	Clicks a map feature using the Info tool	Displays attribute information of the selected feature
7	Views dashboards	Displays KPI and layer summary dashboards based on visible data
8	Reviews information for reporting or decision-making	Provides up-to-date spatial and summary insights
9	Logs out	Ends user session securely

Outcome:

The LGU end user can view maps and dashboards relevant to their city or municipality without modifying data.

B. User Journey Story 2: System Administrator (Admin Role)

Step	Admin Action	System Response
1	Logs in using administrator	Authenticates admin and grants access

	credentials	to admin panel
2	Opens user management module	Displays list of existing users
3	Creates a new user or selects an existing user	Opens user configuration form
4	Assigns a system role (e.g., Viewer, Editor, Admin)	Applies role-based permissions
5	Assigns LGU access restriction (e.g., User1 → City1)	Limits user data access to the assigned city/municipality
6	Saves user configuration	Stores changes and updates access rules
7	Reviews audit logs	Displays recorded user and admin activities
8	Updates, suspends, or revokes access as needed	Enforces updated access rules immediately
9	Logs out	Ends admin session securely

Outcome:

The LGU end user can view maps and dashboards relevant to their city or municipality without modifying data.

SUPERUSER: Sofairich

Login Page: City/ Municipality SELECT >>> LOGIN

C. User Journey Story 3: Owner (Super User)

Item	Action
Permission	• Full access to all LGU datasets and dashboards • User and role management across LGUs • Dataset publishing and configuration • System-wide administrative privileges
Main Flow	

	1. Logs in through the GeoLokal Gateway. 2. Selects any City/Municipality from the LGU selector. 3. Views interactive maps with pan, zoom, layer toggling, and identify tools. 4. Accesses dashboards showing KPIs, CBMS indicators, parcel statistics, and infrastructure summaries. 5. Filters data by barangay and exports reports. 6. Creates, edits, or disables user accounts. 7. Assigns user roles (Viewer/Admin) and LGU access permissions. 8. Uploads and publishes spatial and non-spatial datasets across LGUs. 9. Reviews audit logs and system activity. 10. Switches between LGUs without logging out.
--	---

Outcome:

The Super User is a system-level role with full visibility and control over all LGUs within the Web GIS. This role inherits all Viewer and Admin capabilities and can additionally access datasets, dashboards, and administrative functions across multiple cities and municipalities.

User Story 2 (Admin) and 3 (Super User) **inherit all steps from User Story 1.**

XI. User Interface

Superuser login

GL

GeoLokal

GeoNode-ready LGU portal mock (HTML)

Superadmin

Login as super user, then select a city/municipality to access the LGU login page.

🔑 Demo credentials

Superadmin: admin / admin123

Ibaan: ibaan / ibaan123

Other LGUs: lgu / lgu123

Username

admin

Password

Select City / Municipality

Ibaan, Batangas

Login & Continue

UI prototype only. Leaflet loads online tiles; later replace with GeoNode layers (WMS/WFS).

GL

GeoLokal

GeoNode-ready LGU portal mock (HTML)

LGU Login

Login for the selected LGU. Ibaan uses an orange theme.

🔑 Demo credentials

Ibaan: ibaan / ibaan123

Other LGUs: lgu / lgu123

Username

ibaan

Password

Back

Login to Dashboard

Prototype only. Later connect to GeoNode users/roles (Viewer, Editor, Admin).

Admin Management (Ibaan, Batangas) Pilot

GL

GeoLokal

Ibaan, Batangas

LGU Ibaan, Batangas

User Ibaan Admin

Reset Demo Data

Navigation

Ibaan theme

Dashboard • Users • Audit Logs

Dashboard

User Management

Audit Logs

User Management

Ibaan Admin

Clear Users

Create new username + email. Role is fixed to Viewer (prototype).

Viewer created. Temp password: 7cA8qhTWt7

Username

e.g., ibaan_viewer1

Email

e.g., viewer@ibaan.gov.ph

Role: Viewer

Create Viewer

Username	Email	Role	Temp Password	Created
test	testemail@g.vn	viewer	7cA8qhTWt7	27/01/2026, 08:55:18
123	123@dfs.mm	viewer	6J4KL9HJub	27/01/2026, 08:54:58

Navigation

Ibaan theme

Dashboard • Users • Audit Logs

Dashboard

User Management

Audit Logs

Audit Logs

Ibaan

Clear Logs

Logs are written on: create user, clear users, reset demo.

Timestamp	Actor	Action	Details
27/01/2026, 08:55:18	Ibaan Admin	USER_CREATE	Created viewer test (testemail@g.vn)
27/01/2026, 08:54:58	Ibaan Admin	USER_CREATE	Created viewer 123 (123@dfs.mm)
27/01/2026, 08:54:38	Ibaan Admin	RESET_DEMO	Reset users and audit logs

Dashboard (Ibaan, Batangas) Pilot

GL

GeoLokal

GeoNode-like dashboard mock

LGU Ibaan, Batangas

Logout

Navigation

Ibaan theme

Home • Projects • Maps

Home

Projects

Maps

Quick Links

DRRM

Hazards • critical facilities

Land Use

CLUP • zoning • basemaps

Interactive Map

Leaflet viewer

GeoLokal Dashboard

GeoNode-ready UI

Prototype

A GeoNode-inspired LGU portal structure: authentication → projects → map viewer.

Projects

5 categories

DRRM, Land Use, Revenue, Socio-Economic, Smart City

Maps

Leaflet

Basemaps + demo overlays (replace with GeoNode WMS/WMTS)

Integrations

Planned

External portals/APIs (Geoportal, DRRM sources, RPT portal)

How this maps to GeoNode

- Home = landing + portal description
- Projects = category navigation for datasets/maps
- Maps = GeoNode Map viewer (layers, WMS/WFS)
- Multi-LGU = superadmin selects an LGU; theme/access control change per LGU

GL

GeoLokal

GeoNode-like dashboard mock

LGU Ibaan, Batangas

Logout

Navigation

Ibaan theme

Home • Projects • Maps

Home

Projects

Maps

Quick Links

DRRM

Hazards • critical facilities

Land Use

CLUP • zoning • basemaps

Interactive Map

Leaflet viewer

Projects

Open Map Viewer

Browse categories and sub-menus.

Disaster Risk Reduction Management Mapping

Sub-menus can link to GeoNode groups, layers, or maps.

Hazard Maps

Geohazard Maps

Link →

Volcano and Earthquake Hazard Maps

Link →

Vulnerability and Risk Assessment Maps

Link →

Critical Infrastructure Mapping

Link →

Land Use and Physical Planning

Sub-menus can link to GeoNode groups, layers, or maps.

Land Use & Planning

Comprehensive Land Use Plan Maps

Link →

Zoning Maps

Link →

Topographic and Base Maps

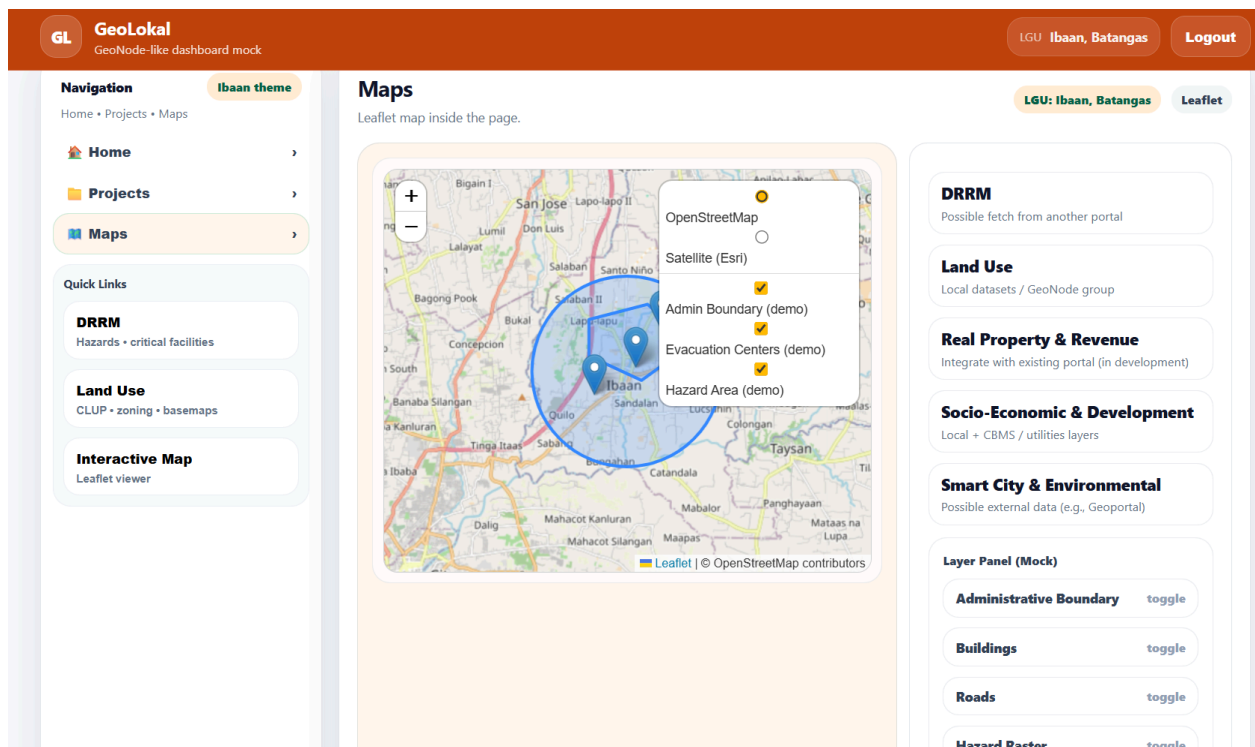
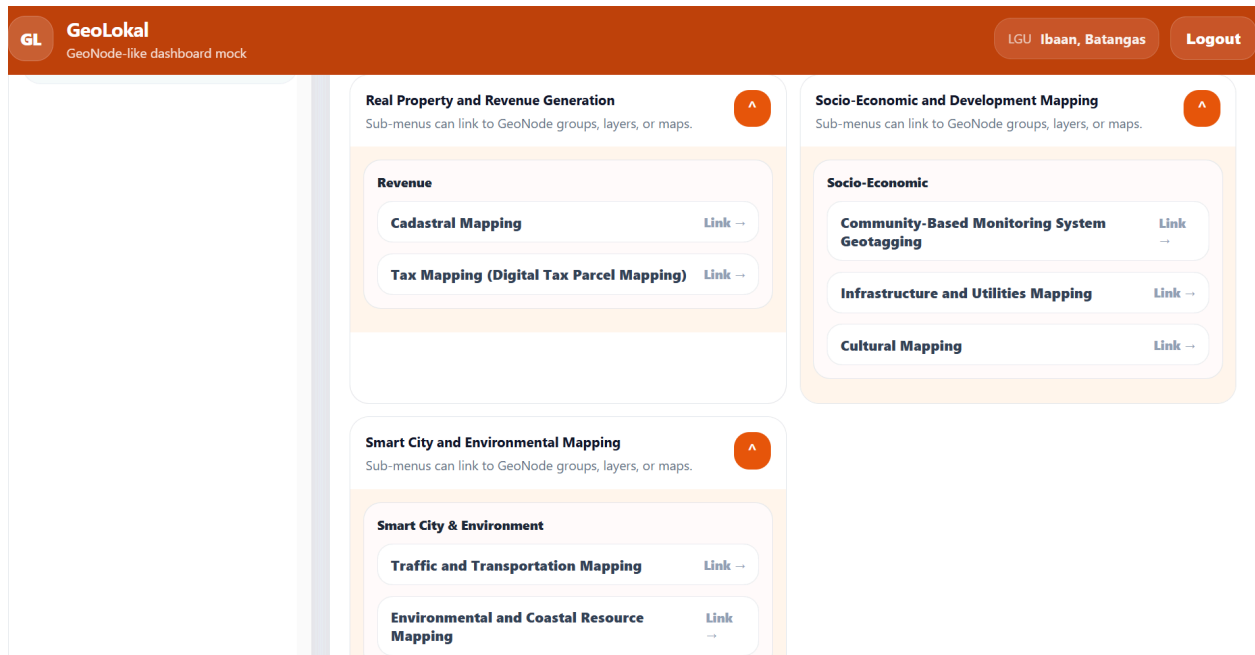
Link →

Real Property and Revenue Generation

Sub-menus can link to GeoNode groups, layers, or maps.

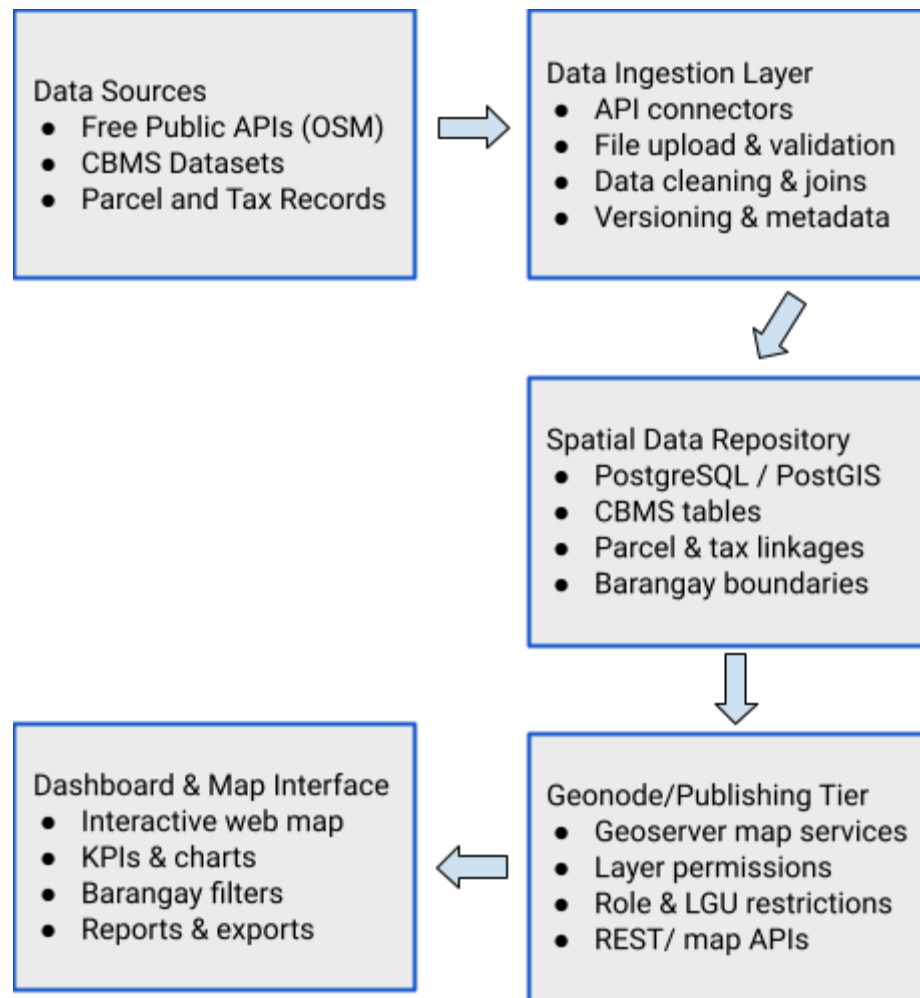
Socio-Economic and Development Mapping

Sub-menus can link to GeoNode groups, layers, or maps.



XII. Data Integration and Connectivity

API (Application Programming Interface) is a standardized method that allows one system to request and receive data from another system in a controlled and secure manner. Instead of manually downloading files or duplicating databases, APIs enable systems to communicate automatically using predefined rules. APIs are commonly used to access maps, datasets, and system functions without exposing internal system details.



A. OpenStreetMap

OpenStreetMap provides free, community-maintained geographic reference data such as roads, buildings, and points of interest that serve as a base layer for map visualization.

B. Geoportal GovPH

Geoportal GovPH is the national spatial data portal that provides authoritative government geospatial datasets and web map services for planning, risk assessment, and inter-agency coordination.

C. CBMS

In-house CBMS supplies barangay-level socio-economic indicators that support evidence-based planning, social development programs, and targeted service delivery.

D. Tax Parcel Mapping

In-house Tax Parcel Mapping integrates parcel boundaries with tax and assessment records to support land administration, revenue management, and spatial analysis of taxation coverage.